Volta Industrial: never be the supplier that stops the OEM line

Live predictive maintenance for aluminum EV battery enclosure lines, on one Databricks platform

Volta Industrial is a Tier-1 automotive supplier. It builds battery trays, covers and cooling plates for three EV makers: Nordvik Motors, Fjord Electric and Aurora Elbil. It ships them just-in-sequence from 12 lines in three plants.

Prepared for **Ingrid Solberg** (COO, executive sponsor) and **Linnea Berg** (Lead App Developer, Volta Digital Manufacturing, who would own and run the app). The app is built for **Bjørn Halvorsen**, Maintenance Supervisor in Plant East.

All companies, people and data are fictional and synthetic. No real customer data was used.

About 20 minutes: 2 setup, 3 problem, 8 demo, 3 value, 4 questions and the ask.

Demo launchpad 1 · the live data journey

Links open in a new tab. Workspace: **fevm-serverless-stable-am1uc2**.

#	Step	Open	What to show
1	Data generation + **Zerobus** ingest	**simulator job** · **pdm_raw.sensor_readings**	Job is running, 96 stations at 1 Hz. Table history: every write is `engineInfo = Zerobus`
2	**SDP** pipeline (continuous)	**SDP pipeline**	Graph: clean → windowed scoring → current state. Data-quality tab: expectations pass/drop
3	Silver and live scores	**pdm_core.sensor_readings_clean** · **pdm_core.station_risk_scores** · **pdm_core.station_health_current**	Fresh rows; Lineage tab from the Zerobus table to the app
4	**ML** model in Unity Catalog	**UC model station_failure_model** · **MLflow experiment** · **training job**	Version 2 = `@champion`, PR-AUC 0.91, lead time per failure mode
5	**Lakebase** sync	**synced-table pipeline** · **pdm_live.station_risk_scores** · **Lakebase project pdm-demo**	Synced table about 4 s behind Delta; work orders as Postgres rows
6	**Databricks App**	**Plant Health Live app** · **app config**	Inject fault on PLT-E-A03 → HIGH → work order → repair → NORMAL. Header shows data age and Lakebase ms
7	**Model Serving** via AI Gateway	**serving endpoint + AI Gateway**	What-if from the app; usage tracking in the gateway

Demo launchpad 2 · AI, governance, code and evidence

#	Step	Open	What to show
8	**Genie** agent	**Genie: Plant Maintenance Agent**	Ask: "Which OEM programs are exposed to a line stop right now?" Benchmarks tab: 12/12
9	Governed KPIs (metric views)	**pdm_ops.maintenance_metrics** · **pdm_ops.station_risk_metrics** · **pdm_ops.oem_delivery_exposure**	One definition per KPI; OEM exposure view joins live risk to JIS contracts
10	**Unity Catalog** governance	**pdm_raw.technicians (masks, row filter)** · **pdm_ops.work_orders_current** · **governance job**	PII masks, plant row filter (app identity sees 6 of 18 rows), tags, grants
11	Industry context	**industry context job**	OEM programs, JIS buffers, line-stop charges, scorecard KPIs computed from the history
12	Code, deployed as one bundle	**GitHub repo** · **databricks.yml** · **src/pipeline** · **src/app**	Everything in Git; deployed code = repo (checksums)
13	Proof, as text	**requirement → evidence matrix** · **evidence/** · **evidence notebooks job** · **end-to-end run log**	Executed notebooks with outputs, raw API responses, timings
14	Deep dive: data quality + CI/CD	**mfg-telemetry-dq-cicd**	Zerobus + Auto Loader, SQL expectations, quarantine, DQ gate, dev/staging/prod promotion

Before demoing: start Lakebase, both pipelines, the simulator and the app; warm the serving endpoint and Genie once.

Who is in the room

	Ingrid Solberg, COO	Linnea Berg, Lead App Developer
Decides	Funds the program and the rollout to all plants	Whether her team can own, extend and run the app in production
Judged on	EBITDA margin under OEM price-downs, OEM charges, the OEM scorecard that decides the next program award, capex	Uptime and response time of the plant apps, release speed, security reviews passed, on-call load
Wants to hear	What it is worth, how fast, what the risk is	How it is built, how data and auth work, how it is tested and deployed, what breaks

The app's user is **Bjørn Halvorsen**, Maintenance Supervisor in Plant East. He is not in the room, but every screen in the demo is his.

The outcome in one slide

Every failure in the test set was flagged before it happened, with a median of about 4 operating hours of warning. Volta's JIS buffers are only 45 to 90 minutes.

	What we showed	For Ingrid	For Bjørn, the app's user
Failures caught early	**271 of 271** on a held-out time window	No OEM line stop from a flagged station	Repairs get planned, not rushed
Warning	**About 4 operating hours** (median)	The OEM never notices	Fix it in the next planned break
Alert precision	**95%** at HIGH	Few wasted inspections	His crew can trust the alerts. About 1 in 20 is a false alarm
Planned repairs	**20% today → about 76%**	Lower downtime and parts cost	Fewer emergency call-outs
Repair minutes per event	**96 → about 48**	More output from the same lines	Half the time per job
Value (illustrative)	**About $3.6M + $4.0M a year**	Own P&L + OEM charges avoided	

In the live run, the at-risk welder was **repaired and never failed**.

The problem, in Volta's numbers

From Volta's synthetic maintenance history (96 stations, 1,096 failures):

	Downtime	Parts
Repair after a failure	**112 min**	**$7,200**
Planned repair	**28 min**	**$870**

- **Only 20% of repairs are planned today.** The rest happen after the station has failed.
- **112 minutes is longer than every JIS buffer** (45 to 90 min). An unplanned failure stops the OEM's line for **about 44 minutes**, which costs **about $595,000** under Volta's contracts.
- **A 28-minute planned repair fits inside every buffer.** The OEM never sees it.
- **And in this niche, the OEM scorecard decides the next vehicle program.** One line stop is a mark against Volta.

Turning a failure into a planned repair turns an OEM line stop into a non-event.

Live: Bjørn's shift on the Aurora Elbil line

▶ **Plant Health Live app** · **Genie: Plant Maintenance Agent**

Station **PLT-E-A03**, the frame welder on line PLT-E-A in Plant East. The line builds battery trays for the Aurora Elbil A3 Hatch, with a 60-minute JIS buffer. 1 demo minute ≈ 1 operating hour.

	Tell	Show	What it means
1 · The alert	"What Bjørn sees on his screen"	The welder starts to overheat. Its risk climbs from 1% to HIGH (79%, then 99%). Top signal: motor current, 10% above normal	Hours of warning against a 60-minute buffer. 95% of these alerts are real
2 · Act and recover	"Now the part that pays"	One click creates a P1 work order, stored in about 12 ms. The technician completes it and the risk returns to NORMAL	A 28-minute planned job instead of 112 minutes. No Aurora Elbil line stop, which would have cost about **$575,000**
3 · Ingrid's question	"The question before an OEM review"	Genie: "Which OEM programs are exposed to a line stop right now?" It ranks the lines by risk-weighted exposure	Ingrid knows her exposure per OEM. Bjørn knows where to send the crew first

Ingrid's view: OEM delivery exposure, live

One governed view joins live risk to each line's OEM contract: "If the riskiest station on this line fails now, does the OEM line stop, and what does it cost?"

Snapshot from the running system (Genie's answer, reproduced in SQL). ▶ **Genie: Plant Maintenance Agent** · **pdm_ops.oem_delivery_exposure**

Line	OEM program	Riskiest station	Risk	OEM stop if it fails	Risk-weighted exposure
PLT-S-B	Fjord Electric FE Pace Sedan	Stamping press	HIGH, 99.7%	51 min	**$913k**
PLT-N-C	Nordvik Motors NV-e5 Crossover	Stamping press	HIGH, 99.9%	51 min	**$609k**
PLT-E-A	Aurora Elbil A3 Hatch	Extrusion CNC	ELEVATED, 48%	55 min	**$294k**

- **Per OEM:** Fjord Electric $1.09M, Nordvik Motors $0.80M, Aurora Elbil $0.46M.
- **Tested before anyone used it:** Genie answered 12 of 12 benchmark questions correctly, against reference SQL.

For Ingrid: what it is worth, and what moves on her scorecard

Illustrative for the 96-station fleet. Assumptions: 2 unplanned failures per station per year, 70% converted into planned repairs, $15,000 per hour of internal line downtime. Volta's finance team replaces these with actuals in the pilot.

Layer	What it is	Per year
1 · Own P&L	About 190 hours of line downtime avoided, plus parts, minus false-alarm inspections	**about $3.6M** (+0.26 OEE availability points)
2 · OEM charges	If 5% of unplanned failures reach the OEM: about 10 stops × $595k, 70% avoided	**about $4.0M** (2%: $1.6M; 10%: $8.0M)
3 · Next program award	A clean delivery scorecard with each OEM	not monetized, decisive

How Ingrid is judged (typical Tier-1, Volta to confirm)	What moves
EBITDA margin after annual OEM price-downs	Layer 1, without more headcount
OEM charges and premium freight	Layer 2
OEM scorecard and next program award	Fewer line stops, exposure per OEM visible live
Capex and free cash flow	More output from the same 12 lines, no new equipment

For Bjørn, the app's user: how his shift changes

Today

- He finds out about a failure when the line stops, and the JIS clock is already running.
- Preventive maintenance follows a calendar, so some stations get serviced too early and others too late.
- Four out of five repairs are emergencies.

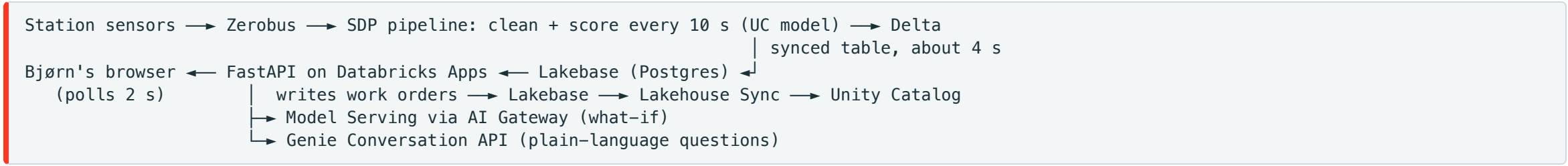
With the app

- **Which station:** the plant screen ranks all stations by failure risk and refreshes every few seconds.
- **Why:** each alert names the top signal, for example motor current 10% above normal, so the technician knows where to look.
- **How much time:** hours of warning, so he schedules the repair into the next planned break.
- **One click:** a work order for the technician, stored transactionally. In the pilot it also flows into the CMMS, so nothing is typed twice.
- **Proof it worked:** after the repair, the station's risk returns to NORMAL on the same screen.

Bjørn's numbers	Today	With 70% of failures caught
Planned share of repairs	20%	**about 76%**
Repair minutes per event	96	**about 48**
Alerts he can trust	none	**95% precision**
Warning before a failure	none	**about 4 operating hours**

For Linnea: how the app is built, and what her team would own

▶ **src/app** · **app config** · **Lakebase project pdm-demo** · **serving endpoint + AI Gateway** · **SDP pipeline** · **databricks.yml** · **end-to-end run log**



Concern	How it works today	Measured
Data quality	Pipeline expectations drop invalid readings and count them. A window with too few readings is not scored	3.9 s sensor to clean data, about 60 s to screen
Reads	The live risk state is a Lakebase synced table: read-only Postgres, primary key (station, window)	4 to 32 ms per query
Writes	Work orders and commands are transactional Postgres rows. Lakehouse Sync streams the changes back to UC	12 to 14 ms round trip
Identity	The app's own service principal. Credentials are injected by the platform; Postgres OAuth tokens are minted and refreshed in the connection pool	no secrets in code
Access	Least-privilege grants on the app identity. A plant-level row filter, proven on that identity	6 → 12 → 6 rows
Model calls	What-if goes through AI Gateway, with every call tracked	41 to 140 ms warm
Deploy	App, pipeline, endpoint and jobs are one bundle in Git. The deployed code is verified against the repo by checksum	one command per environment
Test	A scripted end-to-end run drives every API call through the deployed app, with timings	fault → HIGH → work order → NORMAL

Pilot and the ask

90 days on line PLT-E-A (Aurora Elbil A3 Hatch, Plant East), with Bjørn's crew

Weeks	What happens
1 to 2	Baseline: 12 months of failures, repair times, downtime cost, OEM charges
3 to 5	Connect the line's sensors through Zerobus. Load the CMMS history and the JIS terms
6 to 8	Train on Volta's history. Set the alert threshold with Bjørn
9 to 12	Bjørn's crew runs the alerts in shadow mode, then live

Success, agreed up front: at least 70% of failures flagged with more than 60 minutes of warning; at least 80% alert precision; zero OEM stops from flagged stations; a measured drop in unplanned downtime.

The ask

1. **Ingrid:** sponsor the 90-day pilot on PLT-E-A.
2. **Linnea:** her team co-owns the app from week 1: code, deployment and on-call, with us alongside. Bjørn's crew runs the alerts and sets the threshold with us.
3. **Together:** agree the baseline and the success criteria before day 1.

What we are honest about: the demo uses synthetic data and fictional contracts, so the value model is illustrative until the pilot. About 60 s to screen is right for this decision, but it is not a safety interlock. Alerts name the most deviating sensor, not a full root cause. For Linnea: Genie runs as the app identity, so per-viewer filtering needs on-behalf-of-user auth; the model endpoint scales to zero and its first call after idle took about 60 s; a few settings in the app config are still hard-coded instead of bundle variables.

Appendix: where the numbers come from

Not presented. For questions. All figures are from executed runs, committed as text in this repo. How the industry review gaps were closed: `docs/INDUSTRY_FIT.md`.

Claim	Evidence
271/271 failures flagged, median 244 s lead (about 4 operating hours), PR-AUC 0.91, precision 0.95	`evidence/04_training/02_train_model.md`
112 vs 28 min downtime, $7,200 vs $870 parts, 1,096 corrective vs 268 preventive repairs	`evidence/03_backfill/01_backfill_history.md`
OEM programs, JIS buffers, line-stop charges, exposure per line and per OEM, $595k average charge per failure	`evidence/11_industry_context/05_industry_context.md`
Planned share, repair minutes, parts spend, layer 1 and layer 2 value, OEE points, sensitivity	`evidence/11_industry_context/05_industry_context.md`, sections "Scorecard KPIs", "Value model", "Sensitivity"
Live run on PLT-E-A03 (overheating): fault, HIGH, work order, repair, NORMAL	`evidence/08_app/e2e_fault_injection.md`, `evidence/10_evidence_notebooks/E08_end_to_end.md`
3.9 s to clean data, about 60 s to the screen, 12 ms work orders	`evidence/06_live_pipeline/04_live_evidence.md`, `evidence/08_app/`, `evidence/10_evidence_notebooks/E04_lakebase.md`
Genie 12/12, including OEM exposure questions	`evidence/07_genie/benchmark.md`, `evidence/10_evidence_notebooks/E06_genie.md`
Plant-level access on the app identity (6 → 12 → 6 rows)	`evidence/08_app/governance_as_app_sp.json`, `evidence/10_evidence_notebooks/E05_governance.md`

Layer 1: 96 × 2 = 192 failures. 70% = 134 converted. 134 × 84.6 min = 189 h × $15k = $2.83M. 134 × $6,329 = $0.85M. False alarms ≈ 7 × $7.9k = $0.06M. Net = $3.62M.

Layer 2: 192 × 5% = 9.6 OEM stops. × 70% × $594,853 = $4.0M.

Maintenance KPIs: 70% of 1,096 failures converted: 329 corrective, 1,035 planned. Planned share = 1,035 / 1,364 = 76%. Repair minutes = (329 × 112.3 + 1,035 × 27.7) / 1,364 = 48. Parts = $8.12M → $3.27M.

Volta Industrial, Nordvik Motors, Fjord Electric, Aurora Elbil and all personas are fictional, and so is all of the data.